

Luyao.Derek.Xia

AI Specialist

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My current research focuses on applied AI systems for healthcare, wearable devices, and robotic AI. In the MBIE Smart Ideas project, I work on soft-tissue segmentation and reconstruction, tactile-augmented visual tracking, and mixed-reality surgical guidance. I also develop AI-agent smart glasses and robotic Depth-VLA / Vision-Tactile VLA systems, with experience across hardware deployment, end-to-end dataset construction, perception pipeline development, and vision-language-action algorithm design.

EMPLOYMENT HISTORY

Postdoctoral Fellow, Auckland Bioengineering Institute, University of Auckland

Auckland, New Zealand | Jul 2025 - Present

- Conduct MBIE Smart Ideas research on soft-tissue segmentation and reconstruction, tactile-augmented visual tracking, and mixed-reality surgical guidance for healthcare platforms.
- Develop AI-agent smart glasses and robotic Depth-VLA / Vision-Tactile VLA systems for healthcare, wearable AI, and robotic AI applications.
- Lead hardware deployment, end-to-end dataset construction, perception pipeline development, and vision-language-action algorithm design and development.
- Implement real-time 3D surgical segmentation and tracking pipelines for soft-tissue deformation analysis and intraoperative monitoring.
- Supervise postgraduate students, write research publications and grants, and coordinate academic-industry collaborations.

AI Engineer (Internship), China UnionPay Electronic Payment Research Institute

Shanghai, China | Jun 2019 - Dec 2019

- Participated in an AI-driven non-inductive car payment project integrating computer vision and backend payment systems.
- Developed license plate recognition models for automated vehicle identification and seamless transaction processing.
- Built and deployed a non-inductive payment server using Spring Boot for real-time transaction validation and scalability.
- Integrated AI inference pipelines into production payment workflows, improving automation and reliability.

Data Engineer (Internship), China UnionPay

Shanghai, China | Jul 2018 - Oct 2018

- Designed and implemented database partition strategies to improve data management and system performance.
- Developed and tested database optimization algorithms for query efficiency and scalability.
- Optimized large-scale enterprise databases using MYCAT middleware to support high-concurrency payment operations.

EDUCATION

University of Auckland

Auckland, New Zealand | Visiting PhD Student, Artificial Intelligence in Industry, Faculty of Engineering

Tongji University

China | PhD, Control Science and Engineering, School of Electronic and Information Engineering

- Doctoral Dissertation: Research on Automated Assembly Sequence Planning and Quality Inspection Methods Based on Multi-dimensional Semantic Knowledge Modeling.
- Awards: The First Prize Scholarship; Excellent doctoral dissertation.

Tongji University

China | M.S. degree, Control Engineering, School of Electronic and Information Engineering

- Master dissertation: Research and Application of Production Visualization System based on Augmented Reality in Digital Workshop.
- Award: The Phoenix Scholarship.

SKILLS

Machine Learning | Computer Vision | Graph Learning | Generative AI | Python Programming | AI Agents | Vision-Language-Action Learning | Digital Twin | Augmented and Mixed Reality

PROJECT EXPERIENCE

Multimodal Language Model Enterprise Automation Tool Construction and Deployment

2024.12 - 2025.04 | Supported by Shanghai Zaixing Testing Technology Co., Ltd - Project Lead

- Designed and deployed an enterprise automation tool powered by a multimodal large language model.
- Integrated corporate knowledge and data to build and run end-to-end workflows for business operations.

Huayi Energy & Chemical Digital Twin AI Platform

2021.12 - 2022.09 | Supported by Shanghai Huayi (Group) Company - Major Participant

- Conducted on-site investigations of the Huayi Chemical Plant to assess operational needs.
- Planned functional and technical modules for the digital twin AI platform to optimize factory operations.

Subway Track Area Segmentation and Obstacle Identification

2021.03 - 2023.01 | Supported by Shanghai Qnode Technology Co., Ltd - Project Lead

- Applied 3D point cloud segmentation methods to detect subway track areas.
- Collected and analyzed subway data to build a dataset, and used the SSD algorithm for obstacle identification.

Smart Factory: Technology for Logistics Optimization and Collision Avoidance Based on 3D Real-time Localization

2018.01 - 2022.06 | National Major Program for International S&T; Cooperation: China & Germany - Major Participant

- Managed the overall project and advanced the project schedule.
- Developed a digital twin system for a hydraulic cylinder factory and managed 3D equipment models.
- Led workshop precise positioning technology based on SLAM and designed AR logistics applications.

PUBLICATIONS

1. Liu, Y.*, Xia, L.*, Liu, H., Yan, X., Liang, J., Li, H., & Gai, K. (2026). ALM-MTA: Front-Door Causal Multi-Touch Attribution Method for Creator-Ecosystem Optimization. Proceedings of the Fourteenth International Conference on Learning Representations (ICLR 2026). (Under Review, AI Top Conference, Co-first and Corresponding Author*)
2. Xia, L., Lu, J., Fan, Y., & Zhang, H. Integrating Augmented Reality and Small Object Detection for Enhanced Logistics Management in Flexible Workshops. Computers in Industry (Under Review) (Impact Factor: 9.1, SCI, JCR Q1, Top Journal).
3. Liu, Y., Miao, Y., Xia, L*. Directed Routing Gradient (DRGrad): A Personalized Information Surgery for Multi-Task Learning (MTL) Recommendations. Advances in Neural Information Processing Systems, AAAI 2025 (AI Top Conference, Corresponding Author).
4. Xia, L., Lu, J., Lu, Y., Gao, W., Fan, Y., Xu, Y., & Zhang, H. (2024). Semantic knowledge-driven A-GASeq: A dynamic graph learning approach for assembly sequence optimization. Computers in Industry, 154, 104040. <https://doi.org/10.1016/j.compind.2023.104040>.
5. Xia, L., Lu, J., Lu, Y., Zhang, H., Fan, Y., & Zhang, H. (2024). Augmented reality and indoor positioning based mobile production monitoring system to support workers with human-in-the-loop. Robotics and Computer-Integrated Manufacturing, 86, 102664. <https://doi.org/10.1016/j.rcim.2023.102664>.
6. Xia, L., Lu, J., Zhang, H., Xu, M., & Li, Z. (2022). Construction and application of smart factory digital twin system based on DTME. The International Journal of Advanced Manufacturing Technology, 120(5-6), 4159-4178. <https://doi.org/10.1007/s00170-022-08971-1>.
7. Lu, J., Xia, L*, Zhang, H., & Xu, M. (2022). Research and application of manufacturing enterprise digital twin ecosystem. Computer Integrated Manufacturing Systems, CIMS, 28(8), 2273-2290. <https://doi.org/10.13196/j.cims.2022.08.001>. (EI Journal, Corresponding Author).
8. Xia, L., Lu, J., & Zhang, H. (2020, December). Research on construction method of digital twin workshop based on digital twin engine. In 2020 IEEE International Conference on Advances in Electrical Engineering and Computer Applications (AEECA) (pp. 417-421). IEEE. <https://doi.org/10.1109/AEECA49918.2020.9213649>.

ADDITIONAL INFORMATION

- Software Copyright of Tongji Production Monitoring System Based on Augmented Reality (V1.0). Certificate No.0949442, July 2021. (Issued Software).
- Software Copyright of Tongji Part Assembly Induction Prototype System Based on Augmented Reality (V1.0). Certificate No.4749595, September 2019. (Issued Software).
- Software Copyright of Tongji iOS Experimental Unit Digital Twin System (V1.0). Certificate No.3003770, June 2018. (Issued Software).